

[Temporary Title]From Model to Memory: Medium Alignment in Prosomoiosi (Simulation)

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Abstract

Prosomoiosi (Simulation) is a real-time audiovisual exploration of how successive modelling frameworks overwrite cultural memory. Grounded in media archaeology, the work introduces the concept of *medium alignment*, arguing that ideas must be voiced through media whose operative logic remains visible. A live diffusion pipeline—combining TouchDesigner, StreamDiffusion, and ComfyUI Flux up-scaling—stages the algorithmic politics of selective remembrance and invites audiences to renegotiate authorship at the human-machine frontier.

CCS Concepts

• **Applied computing** → **Media arts**; • **Computing methodologies** → *Artificial intelligence*.

Keywords

generative AI, medium alignment, real-time diffusion, text-to-video synthesis, StreamDiffusion, TouchDesigner, AI film, media archaeology

1 Introduction

Over the past decade, the impact of generative artificial intelligence (AI) on audiovisual media has moved rapidly from research laboratories to the core workflows of the film industry. In 2024 the Academy of Motion Picture Arts and Sciences explicitly allowed films containing AI-generated footage to compete for Oscars—a symbolic step that not only reflects Hollywood's recognition of a technological paradigm shift but also signals a loosening of traditional production standards. In the same year, the Runway AI Film Festival (New York and Los Angeles) and the RAIN Film Fest (Barcelona) were launched, indicating that AI-based image creation is transitioning from peripheral artistic practice to a scaled ecosystem; the inauguration of the first Asian AI Film Festival (HKUST AI Film Festival) in 2025 further confirms the global spread and cultural diversification of this trend.

Technically, breakthroughs in generative models for script writing, storyboarding, image synthesis, and even long-form video generation have opened unprecedented real-time iterative spaces for creators. Text-to-video models such as Runway Gen-2¹, OpenAI's Sora², and Google's Veo³, built on large-scale spatio-temporal diffusion frameworks, produce controllable, high-definition, and narratively coherent output; meanwhile, text-to-image models like

Stable Diffusion⁴, Flux⁵, and Midjourney⁶, together with large language models (LLMs) for script development and world-building, are reshaping the traditional pre-production workflow. Researchers note that AI tools not only increase production efficiency but also introduce a "machine vision" cultural grammar that co-evolves with audience perception.

Yet this technological surge exposes a largely overlooked contradiction: when artists embrace a new medium while retaining an old conceptual framework—or conversely, use new concepts to drive an old medium—a misalignment arises between medium and idea. The consequences are typically:

- (1) The new technology is treated merely as an "accelerator," unable to participate substantively in the work's intent;
- (2) The concept is forced to yield to the tool's default paradigm, resulting in demonstrative or techno-virtuosic superficiality;
- (3) Audiences perceive chiefly a "technological spectacle" rather than the work's thematic core, thereby weakening attention to the artistic statement itself.

This article addresses the phenomenon of "medium/concept misalignment" by taking the moving-image work Prosomoiosi (Simulation) as its central case study. The piece adopts "simulation" as its entry point and, through a series of investigations into the technical, historical, and cultural semantics of simulation, gradually formulates and explicates the notion of "medium alignment": the deliberate re-alignment of a work's conceptual agenda with the operational logic of the medium employed, so that the technical procedure becomes a perceptible part of expression rather than a concealed backstage. The work's technical architecture—in which StreamDiffusion⁷ is invoked inside TouchDesigner⁸ for real-time image generation, augmented by dynamic parameter curves and ComfyUI Flux-based tiled up-scaling—embodies this idea: the diffusion process is rendered visible and incorporated into the narrative, enabling audiences to perceive directly how algorithms shape the image. The paper contends that only under the premise of medium alignment can generative moving images transcend tool-oriented and techno-virtuosic tendencies to become a dynamic arena for renegotiating memory, history, and subjectivity.

¹<http://runwayml.com>

²<https://openai.com/sora/>

³<https://deepmind.google/models/veo/>

⁴<https://huggingface.co/CompVis/stable-diffusion-v1-4>

⁵<https://huggingface.co/black-forest-labs/FLUX.1-dev>

⁶<https://www.midjourney.com/>

⁷<https://github.com/cumulo-autumn/StreamDiffusion>

⁸<https://derivative.ca/>

2 Reference Artworks

To contextualize our research, this section examines a selection of artworks that trace the evolution from early interactive systems to contemporary AI-driven critiques of perception. A foundational example is *A-Volve* (1994) by Christa Sommerer and Laurent Mignonneau, a pioneering installation where visitors' drawings are instantiated as virtual creatures [?]. The work foregrounds themes of artificial life and embodied interaction, positioning the human as a direct creator within a digital ecosystem. Decades later, this focus on perception and digital life has been radically reconfigured by deep learning. Rather than granting users direct creative agency, many contemporary artists use AI to interrogate the very process of seeing. For instance, Memo Akten's *Learning to See* (2017) employs real-time neural networks to reveal how a model reconstructs reality through the biased filter of its training data, explicitly questioning the relationship between perception and truth [?]. A related, though aesthetically distinct, exploration is Scott Eaton's *Entangled II* (2019), which uses a bespoke neural network to process abstract footage into morphing, quasi-human figures, examining the aesthetics of machine perception and the human tendency for pareidolia [?]. Taking this critique of data's influence a step further, *POSTcard Landscapes from Lanzarote* (2022) by Varvara & Mar leverages a StyleGAN2 model trained on distinct datasets to generate two opposing visual realities of the same location: the tourist's view and the local's view [?]. By juxtaposing these AI-generated "gazes," the work offers a powerful commentary on how curated data not only reflects but actively shapes cultural identity and memory, echoing John Urry's concept of the "tourist gaze" [?]. Together, these works illustrate a significant shift from celebrating direct interaction with artificial life to a critical investigation of how AI models, and the data they are fed, mediate and construct our reality.

3 Case Study : Prosomoiosi (Simulation)

3.1 Artwork Description

Prosomoiosi (Simulation) is a video work created through a real-time generative pipeline that, through a media-archaeological lens, advances the notion of "medium alignment" to probe how successive technological regimes continually overwrite memory and identity. The work pursues two mutually reinforcing aims: (1) to unmask—via a stratified narrative of simulation logics—the quiet manner in which each new medium overprints cultural remembrance; and (2) to disclose the algorithmic selectivity and symbolic authority of AI-driven image synthesis by staging a reflexive demonstration built on a TouchDesigner–StreamDiffusion pipeline featuring real-time multi-prompt editing and ComfyUI up-scaling with the flux-dev f16 model. By allowing text prompts, stochastic noise, and live performer input to co-evolve on the exhibition floor, the piece shifts image production from depiction to ontological negotiation, inviting audiences to rethink the future interplay of human creativity and machine simulation.

Simulation here is not mere copying or representation; rather, it is a power-laden rewriting produced through the interplay of technology, archives, and algorithms. Wolfgang Ernst notes that archival temporality shapes memory even as it quietly edits the past [?]; Walter Benjamin anticipated the aura's erosion under mechanical reproduction [?], and Jean Baudrillard later warned of

simulacra supplanting reality [?]. Charting a trajectory "from clay to code"—from sand tables and armillary spheres to deep learning and GANs—modeling technologies have evolved from cognitive tools to arbiters of truth: large networks such as GPT-4 and ESM-2 autonomously conjure worlds from latent space and even guide human decision-making, reshaping the human–reality relation. Within this landscape, video games have become the most pervasive simulation medium; by breaching the "fourth wall," they place players in a hybrid third space where procedural rhetoric lets symbolic capital and ideology permeate interaction, extending McLuhan's dictum that "the medium is the message" [?] alongside Bourdieu's analysis of symbolic power [?]. Educational and political simulations—SimCity, PeaceMaker—as well as Lorenz-style chaotic systems further attest to simulation's profound influence on behaviour and cognition. Faced with the medium mutations of algorithmic culture, art must enact medium alignment, expanding AI alignment into a perceptual-symbolic calibration at the level of the medium itself; situated between Stiegler's "third memory" [?] and Yuk Hui's "cosmic technics," [?] this posture seeks to cultivate a future symbiosis among humans, machines, and media.

3.2 Production Workflow

The real-time generation pipeline of this project consists of four stages—"material preparation → diffusion generation → high-resolution upscaling → post-production," with each stage optimised for live performance.

Material Preparation. Source footage includes public-domain vintage films and live gameplay recordings, uniformly transcoded to 24 fps H.264 in Adobe Premiere Pro. To optimize diffusion processing on RTX 4090 hardware, footage is slowed to 4-14 fps ($0.2\times-0.5\times$ speed) and streamed via NDI to TouchDesigner for real-time generation.

Real-Time Diffusion Generation (TouchDesigner + StreamDiffusion plug-in). Hardware: AMD Ryzen 9 7950X / NVIDIA RTX 4090 (24 GB VRAM). In TouchDesigner, the StreamDiffusion node developed by dotsimulate is loaded. After configuring the project's dependencies, every video frame undergoes image-to-image (img2img) diffusion. Models such as Stable Diffusion XL, Stable Diffusion 1.5, and their LoRAs are switched dynamically to achieve desired artistic effects. StreamDiffusion parameters—Prompt, CFG Scale, Steps, Denoise, etc.—are bound to key-frame curves via the Animation COMP curve editor in TouchDesigner, allowing precise control over all generation parameters. Performers can pre-program or live-edit these curves to sculpt narrative rhythm, achieving smoother and more coherent results. The workflow also integrates ControlNet's HED (Holistically-Nested Edge Detection) branch within TouchDesigner; by parameterizing the edge-strength weight, the diffusion process is further constrained to converge on scene geometry and spatial alignment. Issues encountered: node errors, ghosting, and sporadic artifacts.

High-Resolution Upscaling (ComfyUI Flux Workflow). In this Flux workflow for high-definition image generation, preprocessing and region-wise reconstruction form the core foundation. The overall design aims to ensure generation quality, adapt to different VRAM environments, enhance large-image handling, and improve detail expression. First, the image is imported via a LoadImage node

and uniformly resized to a standard size (e.g., 1024×1024) with an ImageResizeKJ node, ensuring compatibility with downstream modules while controlling interpolation. Next, an UpscaleModelLoader node loads a super-resolution model (e.g., "4x NMKD-Siax 200k"), and the ImageUpscaleWithModel node performs a true 4× enlargement. This step not only increases resolution but also lays a high-dimensional foundation for subsequent generation and refinement, thereby enhancing overall texture and edge sharpness.

After upscaling, to alleviate VRAM pressure from high resolutions and enable multi-tile parallel generation, the system introduces a TTP (Tile-to-Patch)⁹ slicing mechanism. Specifically, the TTP_Tile_image_size node automatically calculates appropriate tile dimensions based on image size and grid count; the TTP_Image_Tile_Batch node then divides the entire image into multiple small blocks and outputs the tile batch, positional data, original image size, and grid structure. Each tile is processed independently and encoded into latent space to fit the Flux model's generation workflow. Once generation is complete, the TTP_Image_Assy node reassembles the tiles according to their positions and the original layout parameters, ultimately reconstructing a complete image with global structural continuity and rich detail. This mechanism effectively balances the trade-off between VRAM and resolution, achieving local generation with global consistency.

Output and Presentation. The 4K image sequence (4096×4096 PNG) produced by the Flux workflow is imported as a whole into Adobe After Effects, maintaining a 24 fps time base. During post-production, plug-ins such as Lumetri Color, Camera Lens Blur, and Optical Flares are used for color grading, depth-of-field simulation, and light-effect compositing. After rendering the master to cinema specs (DCI 4K 4096×2160 / 24 fps / Rec. 709 Gamma 2.4), Topaz Video AI is used for final frame interpolation (Chronos 2× or 4×) and sharpening (Artemis High Quality), producing a final cut suitable both for DCP theatrical projection and continuous playback in exhibition spaces.

3.3 Presentation and Interaction

The piece is presented in two formats: (1) A 4K DCP version (24/50 fps, REC.709, 5.1 surround sound) designed for theatrical or festival screenings; (2) A gallery edition, looped from the same final master, projected in 4K with four-channel audio. Both formats maintain identical color grading and dynamic range, ensuring consistent presentation across black-box theaters and white-cube gallery settings.

4 Discussion

First, Elon Musk's insistence on "reasoning from first principles rather than by analogy"—i.e., reducing a problem to its most irreducible truths before rebuilding a solution [?]—illuminates a blind spot in contemporary media art. Too often, artists choose a medium by analogy: video because predecessors used video, or AI because it is trending. Such analogical thinking risks misalignment between medium and concept, resulting in works whose technological veneer masks conceptual inertia. By contrast, a first-principles approach would begin with the core question of what a piece wishes

to express and only then select (or invent) the medium whose operational logic best embodies that inquiry. Medium alignment, in this sense, is not an aesthetic luxury but a methodological necessity; if the foundational choice of medium is ill-considered, subsequent formal refinements become largely futile.

Second, the necessity—and future potential—of real-time image generation warrants closer scrutiny. Real-time pipelines push the act of "capture" into an ontological negotiation unfolding in the present: parameters, textual prompts and performer interventions co-evolve, exposing the algorithmic contingencies of every frame. Although such immediacy is limited by current hardware throughput and model latency, these very constraints constitute a visible grammar that can be critically mobilised. By contrast, non-real-time generation resembles the analogue darkroom: the temporal distance allows for meticulous post-production, compositing and authorial revision, yet simultaneously obscures the causal chain between algorithm and image. Both approaches offer legitimate creative affordances; the key, again, is to return to first principles and decide whether a work's conceptual agenda is better served by the transparency and performativity of real-time synthesis or by the reflective plasticity of deferred rendering.

The creative community has already begun releasing lightweight open-source toolkits for live diffusion, such as Vadim Epstein's SDFu framework¹⁰. Epstein showcases the tool's use in live performance and video production on his website[?] and in the experimental short film The Poem[?], reminding us that real-time synthesis is not only a technical issue but also a gateway to new viewing relations and modes of authorship.

5 Conclusion

This paper has argued that contemporary generative-AI practice requires more than technical novelty; it demands a principled reconciliation between medium and concept. Building on Musk's call for first-principles reasoning, we re-framed medium choice as a foundational decision rather than a cosmetic afterthought. The notion of medium alignment articulated here positions the operative logic of a medium—its algorithms, temporalities, and interfaces—as an integral layer of meaning production. Through the case study of Prosomoiosi (Simulation) we demonstrated how a TouchDesigner–StreamDiffusion pipeline, augmented by live parameter editing and Flux up-scaling, can turn real-time image synthesis into both method and message. The work exposes the algorithmic contingencies of diffusion while inviting audiences to negotiate authorship in the very moment of visual emergence. Our comparative discussion showed that real-time and non-real-time pipelines offer distinct affordances—immediacy and performativity versus reflective plasticity—and that the critical task is to select the approach that most faithfully embodies a project's conceptual agenda. Medium alignment thus emerges as a transferable design heuristic for artists and researchers navigating an era in which technical frameworks evolve faster than critical discourse. By treating first-principles reasoning and real-time generativity not as ends in themselves but as instruments of conceptual clarity, future practices can move beyond tool-centred spectacle toward works that disclose, rather than disguise, the politics of their own making.

⁹https://github.com/TTPlanetPig/Comfyui_TTP_Toolset

¹⁰GitHubrepository:<https://github.com/eps696/SDFu>

Code availability and demo

The complete ComfyUI workflow used in this study (workflow/Flux Upscaling workflow.json) is openly available on GitHub at <https://github.com/UninstallAll/MAAPIIcursor>, allowing readers to reproduce and extend our results.

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